Lecture 5: Atmospheres I
Composition, Structure, & Energy Balance

Planetary Systems

Mara Attia

Kapteyn Astronomical Institute
University of Groningen

2026

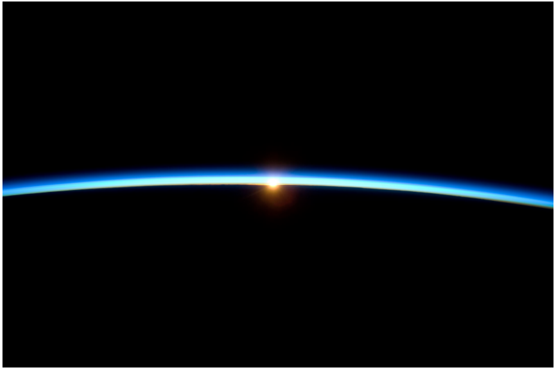
Earth’s thin blue line, ISS STS-129 (2009). Credit: NASA

Learning objectives

By the end of this lecture, you will be able to:

- Classify atmospheric types (primary, secondary, tertiary)
- Derive pressure profiles from hydrostatic equilibrium
- Compute the atmospheric scale height and apply it across the solar system
- Explain how the greenhouse effect raises surface temperature above T_{eff}
- Evaluate atmospheric escape mechanisms and retention criteria

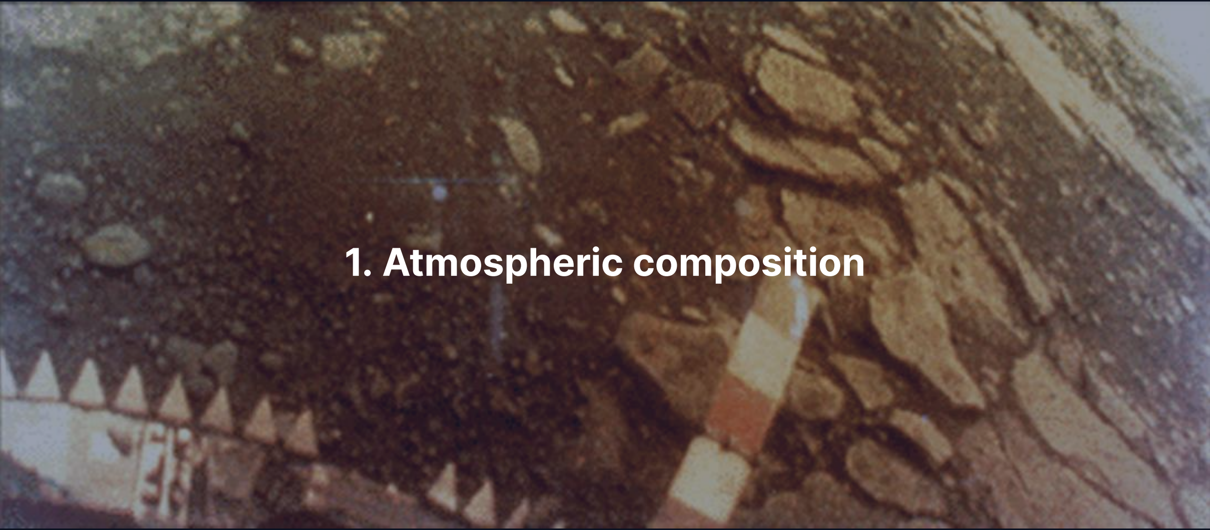
Recap: Lecture 4



Credit: NASA/ISS Expedition 21 crew, public domain

- Metal-silicate separation in magma oceans drove core formation within ~30–60 Myr
- Dynamo requires conducting fluid, convection, and magnetic Reynolds number $\mathrm{Rm} \gtrsim 10$–100
- Mars lost its dynamo between ~4.1 and 3.7 Ga, triggering atmospheric erosion

Today: The thin gaseous envelope that controls a planet's surface conditions.

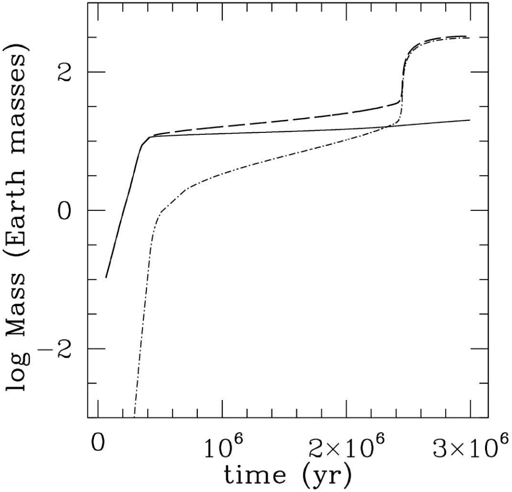


The surface of Venus under its 92-bar atmosphere, Venera 13 (1982). Credit: USSR/NASA NSSDC, public domain

Primary atmospheres

Captured directly from the protoplanetary disk.

- Dominated by $\mathrm{H_2}$ and He (solar composition)
- Requires core mass ≳ 5–10 $M_\oplus$ before disk dispersal in ~3–10 Myr
- **Gas giants:** Jupiter and Saturn retain massive $\mathrm{H_2}$/He envelopes
- **Ice giants:** Uranus and Neptune envelopes are only 10–20% of total mass
- Terrestrial planets are too small to retain primordial $\mathrm{H_2}$/He

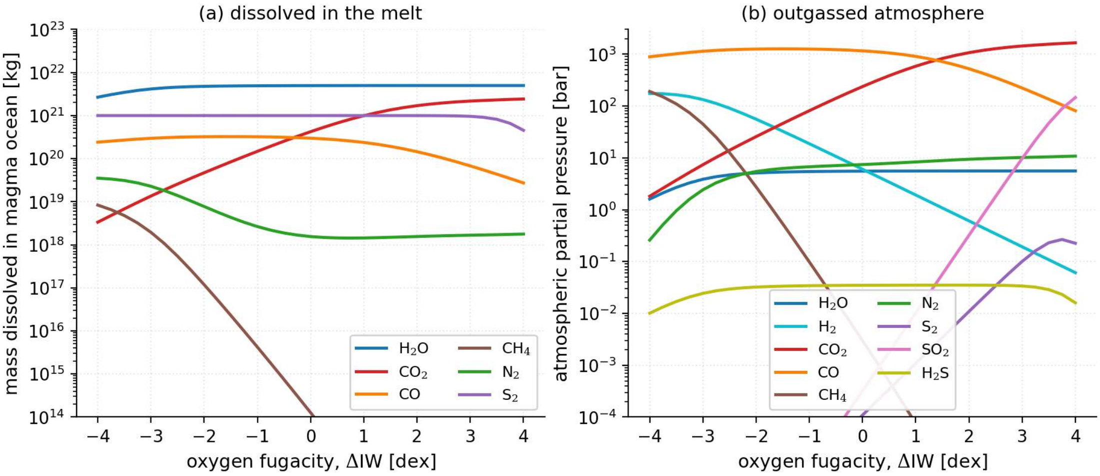


Growth of a giant planet by core accretion: runaway gas accretion builds the H_2/He envelope within the disk lifetime once envelope mass overtakes core mass. Reproduced from Helled et al. (2014), Fig. 1; simulation from Lissauer et al. (2009).

Secondary atmospheres

Produced by outgassing from the interior.

- Magma ocean degassing and volcanism release dissolved volatiles
- Speciation depends on mantle **oxygen fugacity** (f_{O_2} redox state):
  - Oxidising: CO_2, H_2O, N_2
  - Reducing: H_2, CO, N_2
- **Venus:** CO_2-dominated (96.5%), 92 bar
- **Mars:** CO_2-dominated (95%), 6 mbar
- **Titan:** thick N_2 atmosphere (from NH_3), 1.5 bar



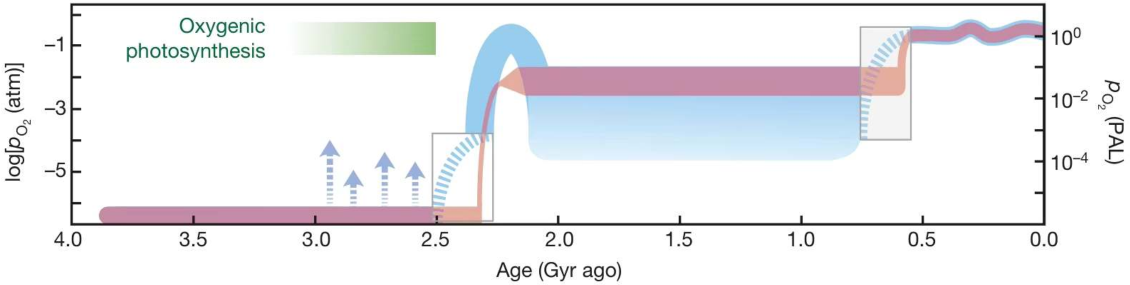
Speciation of an outgassed atmosphere as a function of melt oxygen fugacity: CO, H_2, and CH_4 dominate under reducing conditions, while CO_2 and H_2O dominate under oxidising conditions. Course figure; model of Nicholls et al. (2024).

Tertiary atmospheres

Earth's atmosphere is a tertiary atmosphere produced by billions of years of biological modification.

- Original outgassed inventory was dominated by $\mathrm{CO_2}$ and $\mathrm{N_2}$, similar to Venus
- Oxygenic photosynthesis added the $\mathrm{O_2}$ (21%); the $\mathrm{N_2}$ (78%) is outgassed and accumulated
- Atmospheric $\mathrm{O_2}$ is entirely biogenic and requires active life to persist

Earth's $\mathrm{O_2}$, and so its tertiary character, is continuously maintained by life.

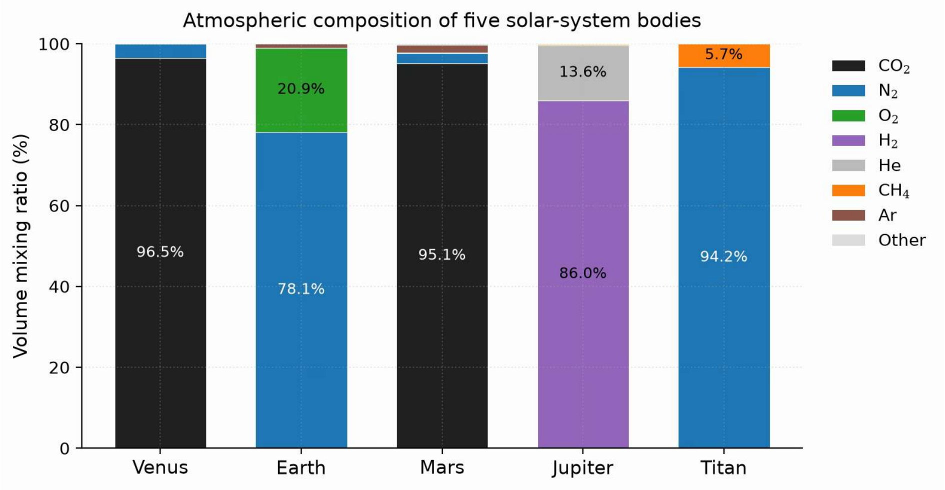


Earth's atmospheric oxygen through time: anoxic for its first ~2 Gyr, rising abruptly at the Great Oxidation Event near 2.4 Ga, and reaching modern levels in the Phanerozoic through biological photosynthesis. Reproduced from Lyons et al. (2014).

Comparative atmospheric properties

	Venus	Earth	Mars	Jupiter	Titan
P_{surf} (bar)	92	1.0	0.006	n/a	1.5
T_{surf} (K)	737	288	215	n/a	94
Dominant gas	$\mathrm{CO_2}$	$\mathrm{N_2}$	$\mathrm{CO_2}$	$\mathrm{H_2}$	$\mathrm{N_2}$
Secondary gas	$\mathrm{N_2}$	$\mathrm{O_2}$	$\mathrm{N_2}$	He	$\mathrm{CH_4}$
μ	43.4	28.97	43.3	2.2	28.6
Type	2°	3°	2°	1°	2°

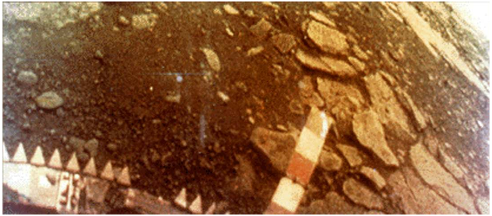
Staggering diversity: surface pressures spanning five orders of magnitude, temperatures from 94 to 737 K.



Atmospheric composition by volume for five solar-system bodies: three origins yield five distinct outcomes across the solar system. Course figure.

Venus: a secondary atmosphere in extremis

A massive secondary atmosphere transforms planetary surface conditions to extreme pressures and temperatures.



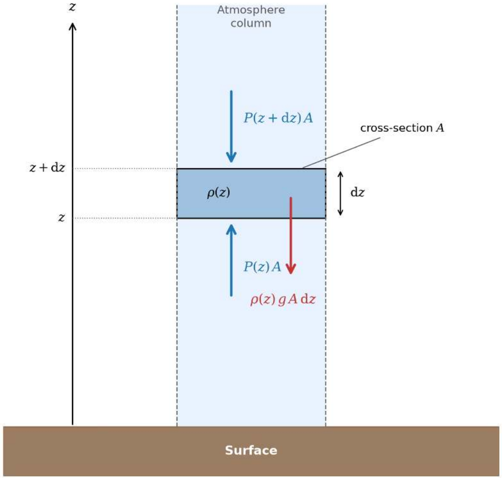
Credit: USSR/NASA NSSDC, public domain

- *Venera 13* landed 1 March 1982 and survived 127 minutes
- Surface pressure of 92 bar and surface temperature of 737 K
- Dense $\mathrm{CO_2}$ atmosphere and cloud deck filter sunlight to a dim orange glow

2. Hydrostatic equilibrium

The thin atmosphere of Mars above its limb, Viking 1 (1976). Credit: NASA/JPL

Hydrostatic equilibrium



Credit: Course figure

Net vertical force on a thin slab of area A and thickness $\mathrm{d}z$ vanishes:

$$P(z)A - P(z + \mathrm{d}z)A = \rho\, g\, A\, \mathrm{d}z$$

$$\boxed{\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g}$$

Pressure decreases upward because each layer supports the weight of all gas above it.

The ideal gas law and barometric formula

The ideal gas law relates density to pressure: $P = \rho\, k_\mathrm{B} T / (\mu\, m_u)$.

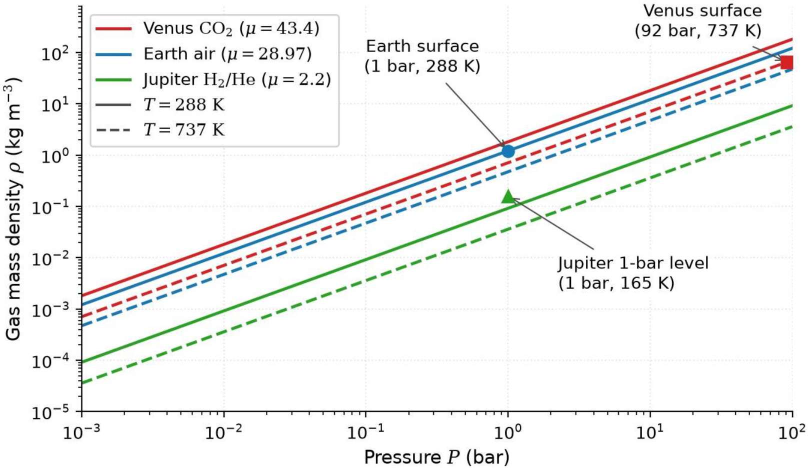
Substituting $\rho(P)$ into hydrostatic equilibrium $\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g$:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\frac{\mu\, m_u\, g}{k_\mathrm{B} T} P = -\frac{P}{H}$$

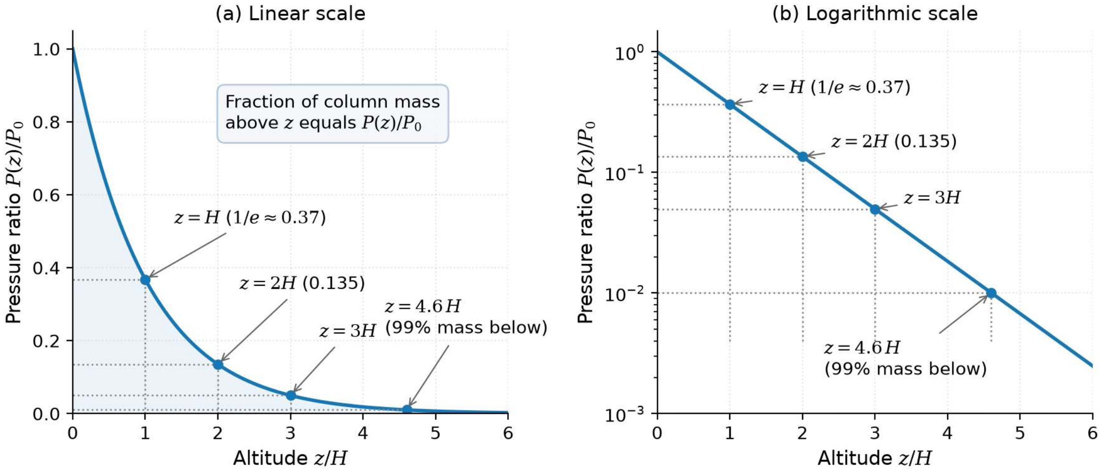
For an **isothermal** atmosphere (constant T), integrating yields:

$$\boxed{P(z) = P_0 \exp\left(-\frac{z}{H}\right)}$$

where $H = k_\mathrm{B} T / (\mu\, m_u\, g)$; pressure drops by $e \approx 2.718$ each scale height.



Ideal gas law in atmospheric form: $\rho = P\mu m_u/(k_B T)$. Density scales with pressure at fixed T; heavier or colder gas is denser. Course figure.



The barometric formula $P/P_0 = e^{-z/H}$: a factor e per scale height, 99% of the column mass below 4.6 H; a straight line on a logarithmic axis. Course figure.

3. Blackboard derivation: scale height

Sunrise over Earth’s limb from orbit. Credit: NASA

Deriving the scale height

Goal: Derive $H = k_{\mathrm{B}} T / (\mu\, m_u\, g)$ from hydrostatic equilibrium and the ideal gas law.

Start from:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g \qquad \text{and} \qquad P = \frac{\rho\, k_{\mathrm{B}} T}{\mu\, m_u}$$

Express ρ in terms of P:

$$\rho = \frac{P\, \mu\, m_u}{k_{\mathrm{B}} T}$$

Substitute:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\frac{\mu\, m_u\, g}{k_{\mathrm{B}} T} P \equiv -\frac{P}{H}$$

The atmospheric scale height

$$\boxed{H = \frac{k_{\mathrm{B}} T}{\mu\, m_u\, g}}$$

Physical interpretation:

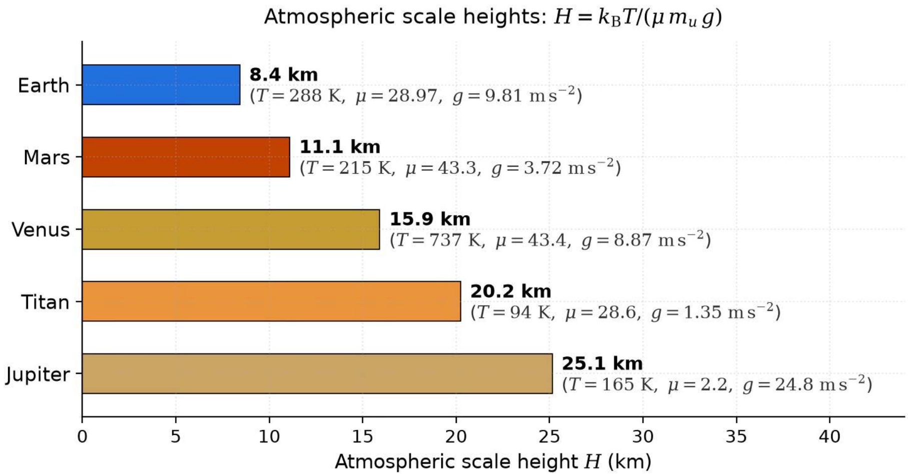
- **Higher** T $\rightarrow$ larger H: hotter gas extends further
- **Heavier molecules** (larger μ) $\rightarrow$ smaller H: harder to loft
- **Stronger gravity** g $\rightarrow$ smaller H: more compressed

Key insight: H encodes the competition between thermal energy and gravitational binding; integrated over the whole column, the surface pressure is the column weight, $P_0 = m_{\mathrm{col}}\, g$.

Scale heights across the solar system

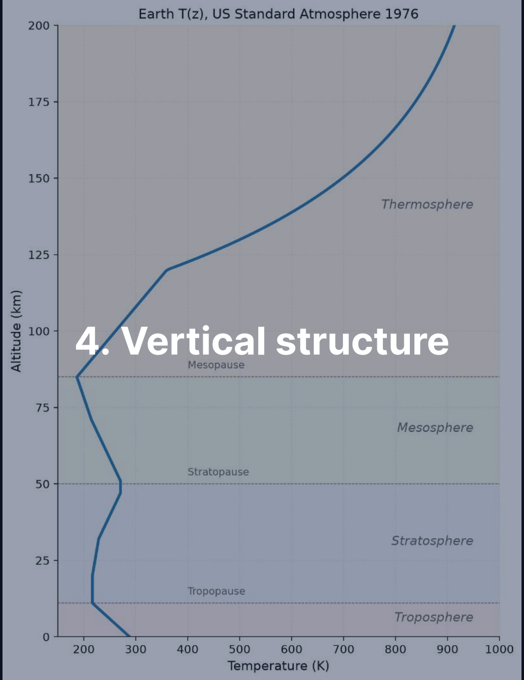
Body	T (K)	μ	g (m s^{-2})	H (km)
Venus	737	43.4	8.87	15.9
Earth	288	28.97	9.81	8.4
Mars	215	43.3	3.72	11.1
Jupiter	165	2.2	24.8	25
Titan	94	28.6	1.35	20

- Jupiter: large H despite strong gravity, because $\mathrm{H_2}$ is very light (μ = 2.2)
- Titan: large H despite low T, thanks to very weak gravity (g = 1.35 m s^{-2})
- Venus: hot but heavy molecules and decent gravity yield moderate H



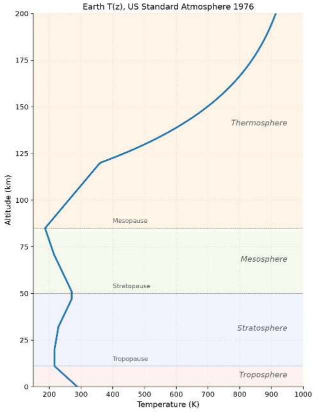
Scale heights recomputed from T, μ and g: light gas (Jupiter) and weak gravity (Titan) give the largest H, Earth the smallest. Course figure.

4. Vertical structure



Earth's atmospheric layers. Course figure

Atmospheric layers



Credit: Course figure; US Standard Atmosphere 1976

Layers defined by the sign of $\frac{\mathrm{d}T}{\mathrm{d}z}$:

1. **Troposphere** (0–12 km): T decreases with z; convective
2. **Stratosphere** (12–50 km): T increases; UV ozone heating
3. **Mesosphere** (50–85 km): T decreases to mesopause (~190 K)
4. **Thermosphere** (>85 km): T increases; EUV heating (T > 1000 K)

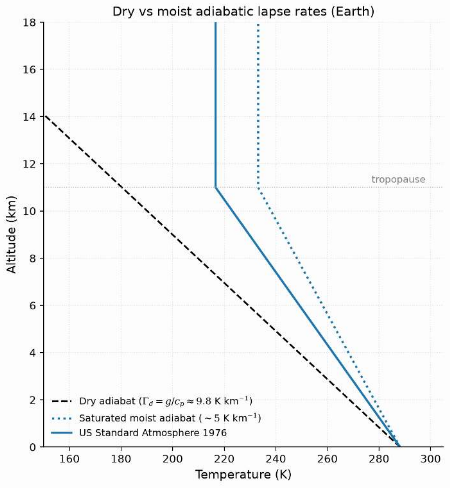
The adiabatic lapse rate

In the troposphere, convection sets the temperature gradient:

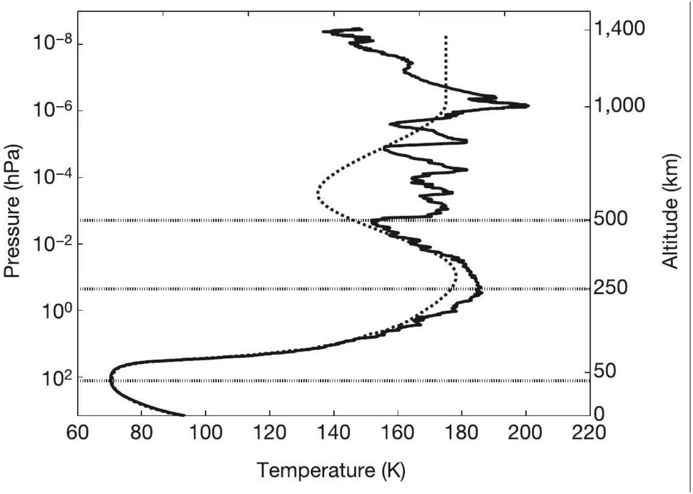
$$\Gamma_d = -\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{g}{c_p}$$

- **Dry adiabatic lapse rate** for Earth: $\Gamma_d = 9.81/1004 \approx 9.8\ \mathrm{K\,km^{-1}}$
- Observed average: ~6.5 $\mathrm{K\,km^{-1}}$ (lower due to latent heat release)
- Physical picture: rising air expands, does work on surroundings, cools

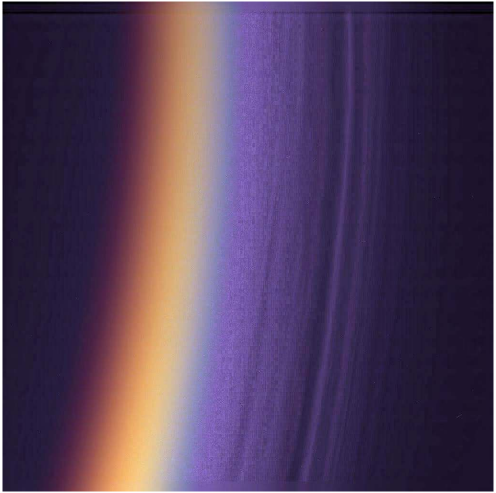
Tropopause height: ~8 km (poles) to ~17 km (equator).



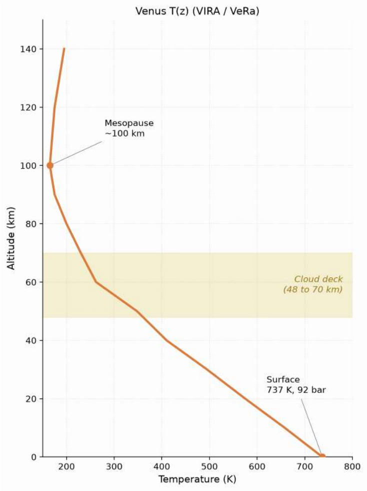
Dry and moist adiabats against the US Standard Atmosphere 1976 reference profile: latent heat makes the moist adiabat shallower, and real convection sits between both limits. Course figure.



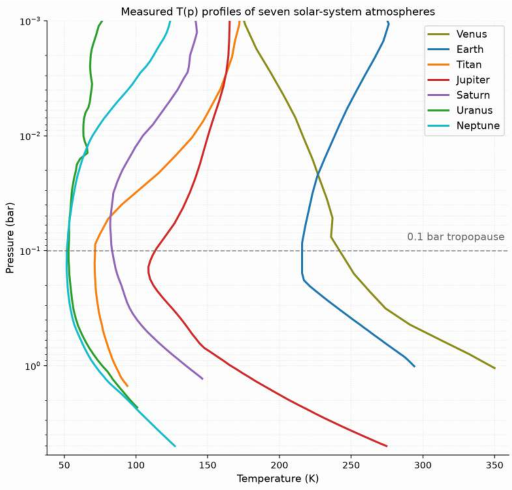
Titan's temperature profile from the Huygens HASI descent: an in situ descent measured the 94 K surface, 0.1 bar tropopause, and stratospheric structure. Fulchignoni et al. (2005).



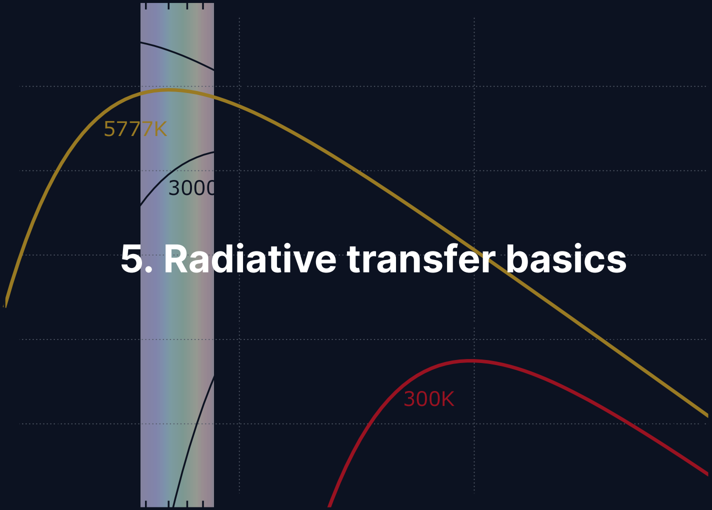
Detached haze layers in Titan's upper atmosphere imaged at the limb by Cassini ISS: photochemical haze layers extending up to 500 km altitude absorb solar UV and heat the stratosphere. Credit: NASA/JPL/Space Science Institute, public domain.



Venus temperature profile, Pioneer Venus/VIRA and Venus Express: without ozone, temperature falls monotonically from 737 K at the surface to the mesopause, reaching 1 bar near 50 km. Course figure.



Measured temperature-pressure profiles of seven solar-system atmospheres: thick atmospheres transition from convective to radiative near 0.1 bar, setting a universal tropopause minimum. Digitized from Robinson & Catling (2014), Fig. 1.



5. Radiative transfer basics

Blackbody spectra. Credit: Wikimedia, public domain

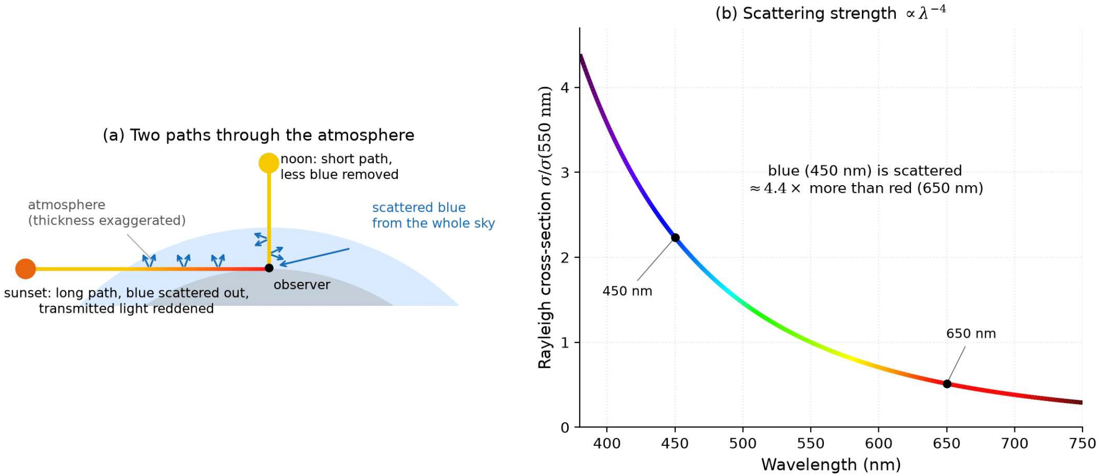
Three interactions: radiation and matter

When radiation passes through an atmosphere:

Absorption: Photon energy converted to internal molecular energy
Key absorbers: CO_2, H_2O, CH_4, O_3, N_2O

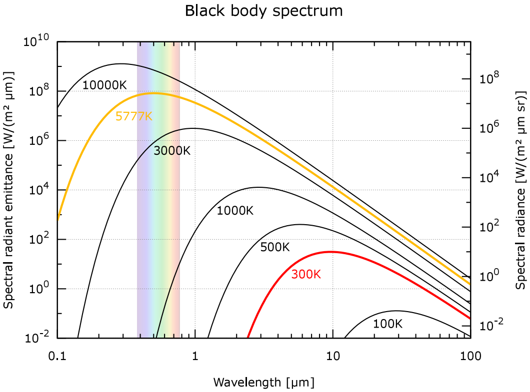
Emission: Warm gas radiates at absorbed wavelengths (Kirchhoff’s law)
Radiates energy directly to space

Scattering: Photons redirected without absorption
Rayleigh scattering ($\propto \lambda^{-4}$) and Mie scattering



Rayleigh scattering and the colour of the sky: shorter path lengths at noon preserve blue scattered light, while longer slant paths at sunset transmit only reddened light; cross-section scales as λ^{-4}. Course figure.

Stellar vs. planetary emission



Credit: Wikimedia, public domain

- Sun (~5800 K): peak at ~0.5 μm (visible)
- Planet (~300 K): peak at ~10 μm (thermal IR)

This **wavelength separation** is the physical basis of the greenhouse effect: the atmosphere can be transparent at visible wavelengths but opaque in the infrared.

Optical depth and the Beer-Lambert law

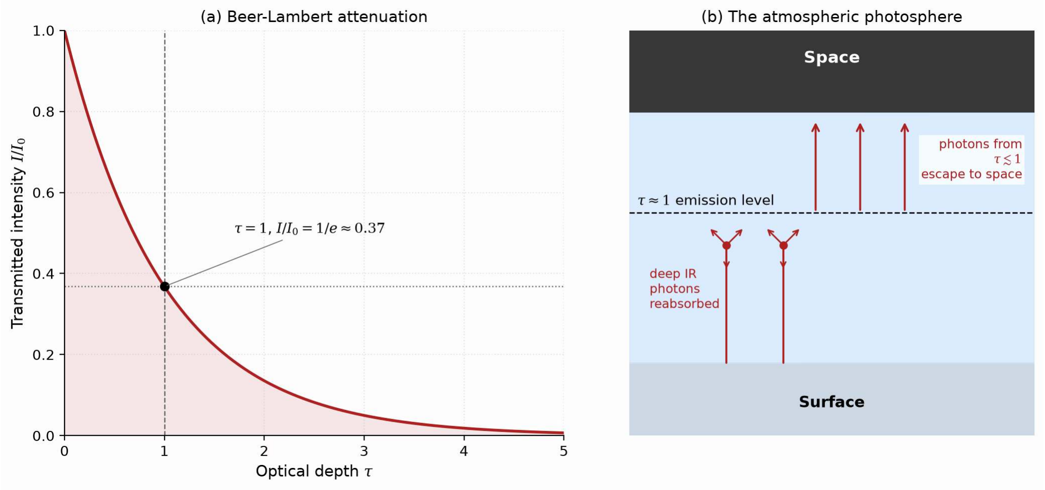
Optical depth τ measures cumulative absorption along a path:

$$\tau = \int_0^s \kappa \, \rho \, \mathrm{d}s'$$

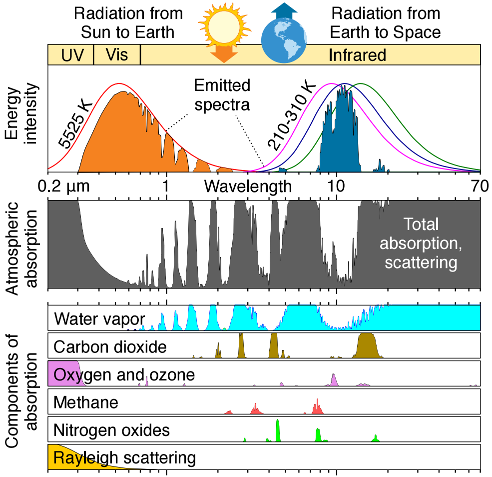
Beam intensity decays following the **Beer-Lambert law**:

$$\boxed{I = I_0 \, e^{-\tau}}$$

- **Optically thin** ($\tau \ll 1$): radiation passes freely through the gas
- **Optically thick** ($\tau \gg 1$): photons are repeatedly absorbed and re-emitted
- **Photosphere** ($\tau \approx 1$): thermal radiation escapes directly to space

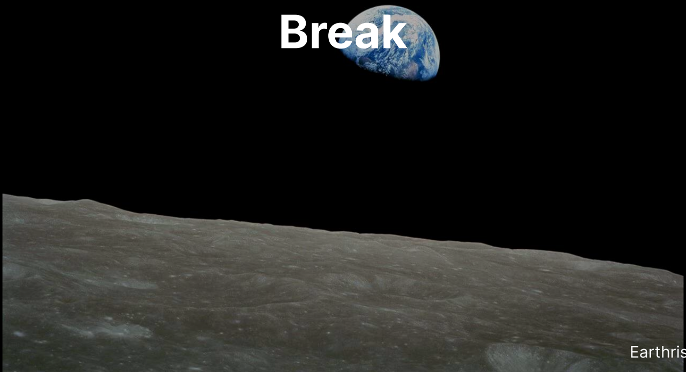


The atmospheric photosphere: thermal radiation escapes to space from the $\tau \approx 1$ level, setting the effective temperature. Course figure.



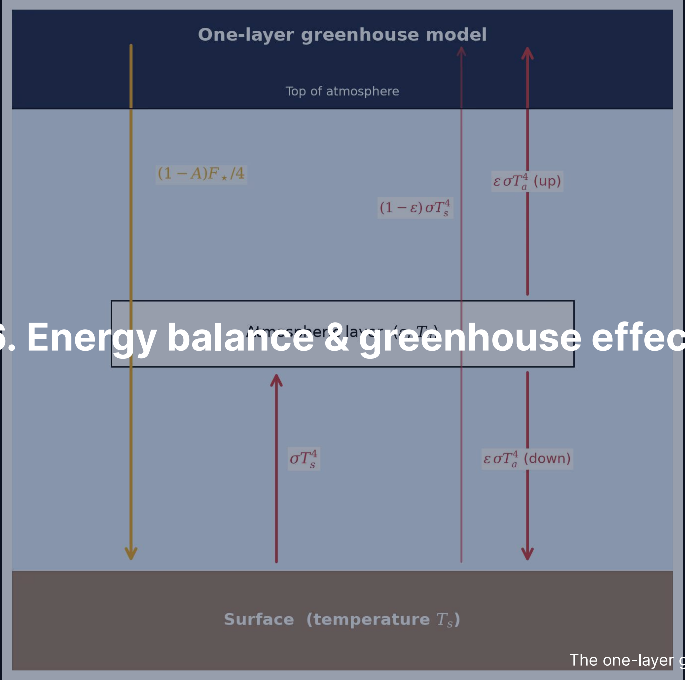
Earth's atmospheric absorption spectrum from UV to IR: greenhouse gases absorb outgoing surface infrared while leaving visible sunlight and the 8–13 μm window transparent. Credit: Wikimedia Commons, public domain.

Break



Earthrise, Apollo 8 (1968). Credit: NASA

6. Energy balance & greenhouse effect

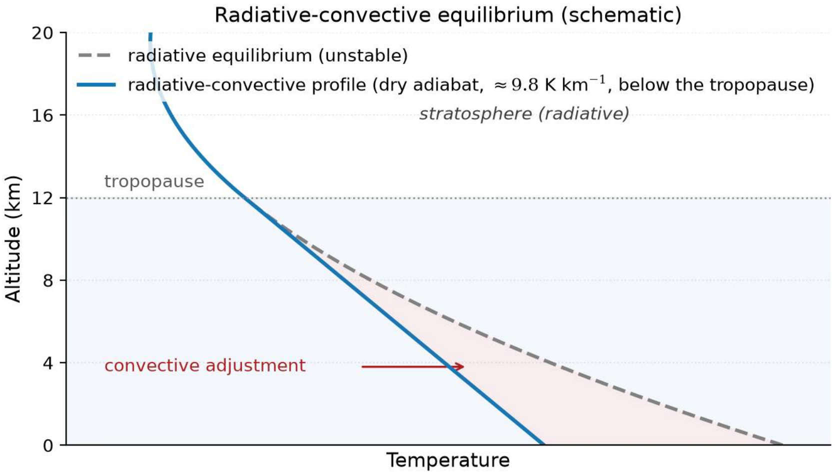


The one-layer greenhouse model. Course figure

Radiative-convective equilibrium

Atmospheres divide into a convective lower layer and a radiatively controlled upper layer.

- **Troposphere**: greenhouse opacity drives convective instability along the adiabat
- **Stratosphere**: optically thin gas reaches radiative equilibrium, stable against convection
- **Tropopause**: boundary where the radiative temperature gradient becomes shallower than the adiabat



Radiative-convective equilibrium (schematic): the radiative profile is superadiabatic near the surface, convection holds the troposphere on the dry adiabat, the tropopause is where the two gradients cross. Course figure.

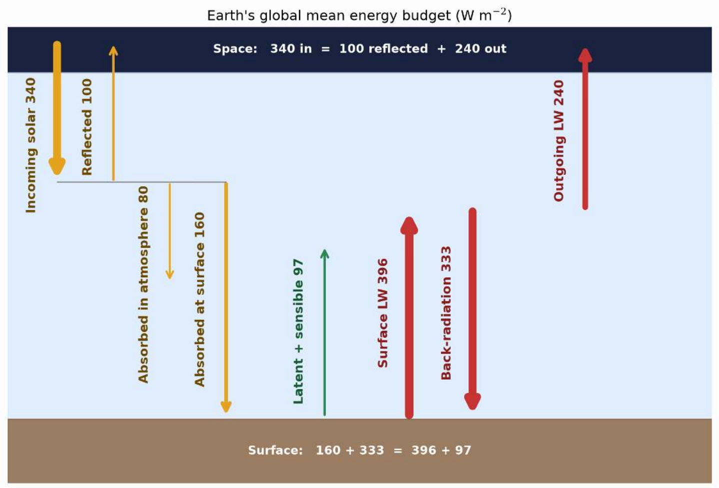
Planetary energy balance

Absorbed stellar power equals emitted thermal radiation:

$$(1 - A)\frac{S}{4} = \sigma T_{\mathrm{eff}}^4 \qquad \text{where} \quad S = \frac{L_\star}{4\pi d^2}$$

- A: Bond albedo; S: stellar flux ($S_0 = 1361\ \mathrm{W\,m^{-2}}$ at Earth)
- Factor $1/4$: ratio of intercepting cross-section πR_p^2 to emitting sphere $4\pi R_p^2$

$$\boxed{T_{\mathrm{eff}} = \left[\frac{(1 - A)\,L_\star}{16\pi\sigma\,d^2}\right]^{1/4} = \left[\frac{(1 - A)\,S}{4\sigma}\right]^{1/4}}$$



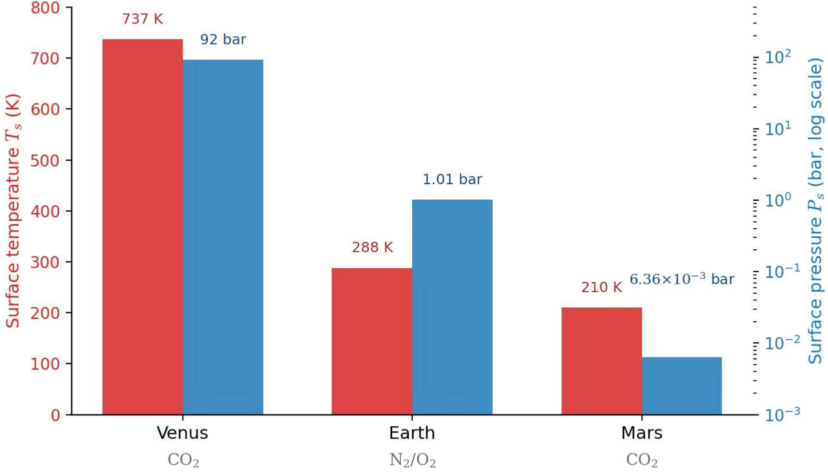
Earth's globally averaged energy budget in W m^{-2}: atmospheric back-radiation of 333 W m^{-2} supplies more energy to the surface than direct sunlight (160 W m^{-2}). After Trenberth et al. (2009).

Effective vs. surface temperature

Body	A	d (AU)	T_{eff} (K)	T_{surf} (K)	ΔT (K)
Venus	0.77	0.72	227	737	+510
Earth	0.30	1.00	255	288	+33
Mars	0.25	1.52	210	215	+5
Jupiter	0.34	5.20	110	165*	+55

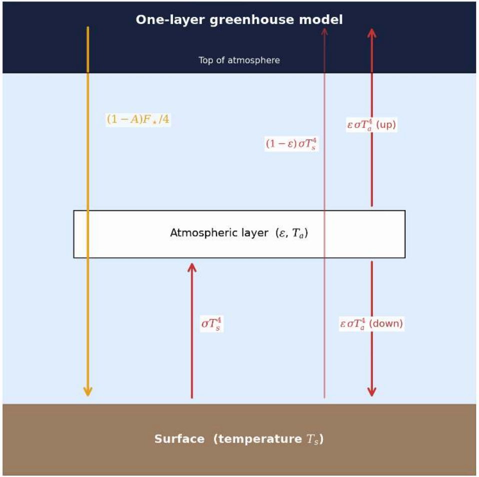
*Jupiter 1-bar level; excess partly from internal heat (Lecture 3).

$\Delta T = T_{\text{surf}} - T_{\text{eff}}$ reveals the **greenhouse effect**.
Venus: 510 K. Earth: 33 K. Mars: ~5 K (only 6 mbar, no water-vapour amplifier).



Surface temperature and surface pressure of Venus, Earth, and Mars: planets of similar bulk composition span a factor of 10^4 in surface pressure and 500 K in surface temperature due to greenhouse warming. Course figure; data from NASA.

The greenhouse mechanism



Credit: Course figure

1. Sunlight (visible, ~0.5 μm) reaches the surface
2. Surface emits thermal IR (~10–15 μm)
3. Greenhouse gases absorb outgoing IR
4. Absorbing layer re-emits: half up, half down
5. Downward emission adds energy to surface

Result: $T_{\mathrm{surf}} > T_{\mathrm{eff}}$

One-layer greenhouse model

Single atmospheric layer with infrared emissivity ε, transparent to visible sunlight:

Flux balance:

$$\text{Atmosphere:} \quad \varepsilon\sigma T_s^4 = 2\varepsilon\sigma T_a^4$$

$$\text{Surface:} \quad (1 - A)\frac{F_\star}{4} + \varepsilon\sigma T_a^4 = \sigma T_s^4$$

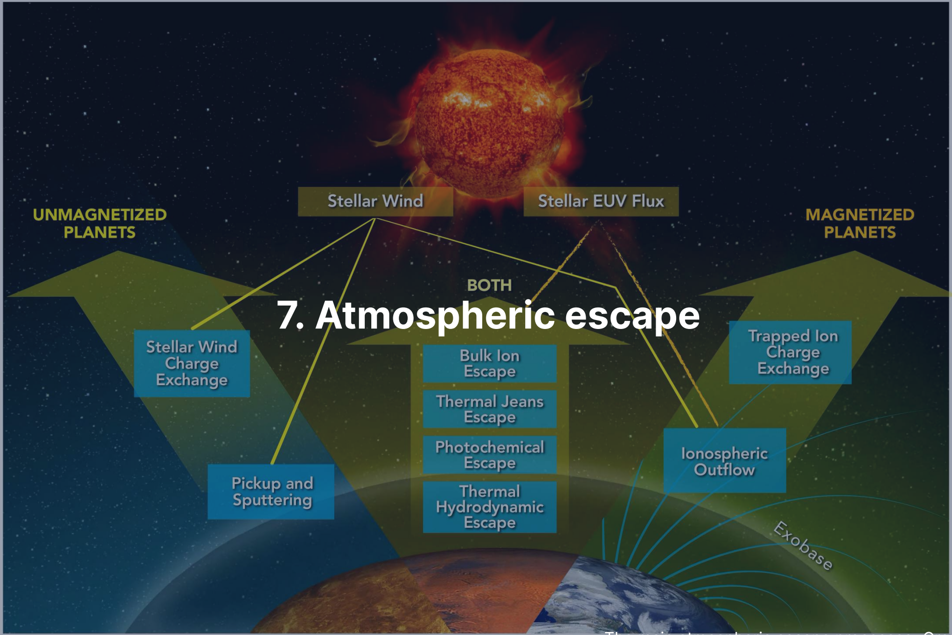
$$\text{TOA:} \quad (1 - A)\frac{F_\star}{4} = \sigma T_{\mathrm{eff}}^4$$

Surface temperature:

$$\boxed{T_s = T_{\mathrm{eff}}\left(\frac{2}{2 - \varepsilon}\right)^{1/4}}$$

- $\varepsilon = 0$ (transparent): $T_s = T_{\mathrm{eff}}$
- $\varepsilon = 1$ (opaque): $T_s = 2^{1/4}\,T_{\mathrm{eff}} \approx 1.19\,T_{\mathrm{eff}}$ (Earth: ≈303 K)

7. Atmospheric escape



The main atmospheric escape processes. Gronoff et al. (2020)

Thermal (Jeans) escape

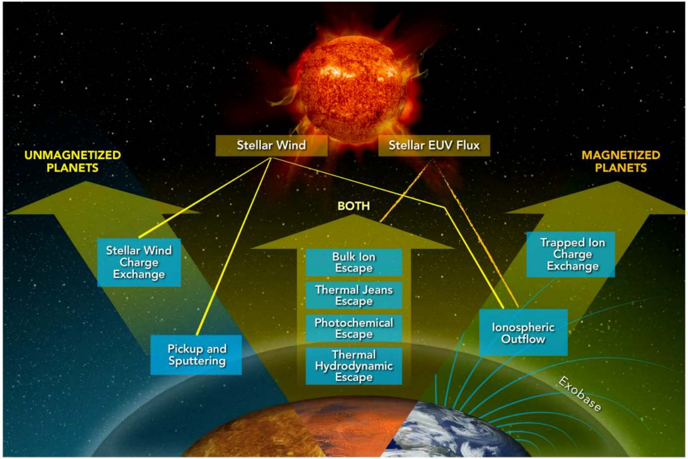
Molecules follow a Maxwell-Boltzmann distribution whose high-velocity tail can exceed escape speed.

The **Jeans escape parameter** at the exobase compares gravitational to thermal energy:

$$\lambda_J = \frac{G M_p\, m}{k_\mathrm{B} T_\mathrm{exo}\, r_\mathrm{exo}} = \frac{v_\mathrm{esc}^2}{v_\mathrm{th}^2}$$

with $v_\mathrm{th} = \sqrt{2 k_\mathrm{B} T_\mathrm{exo} / m}$ the most probable thermal speed.

- $\lambda_J \gg 1$: escape is negligible because the tail is depleted
- $\lambda_J \lesssim 2$–3: thermal escape drives rapid atmospheric loss
- Escape flux $\Phi_J \propto (1 + \lambda_J)\, e^{-\lambda_J}$ depends exponentially on λ_J



The main atmospheric escape processes: thermal, photochemical, and ion channels operate on all planets, while planetary magnetic fields select which pathways dominate. Gronoff et al. (2020).

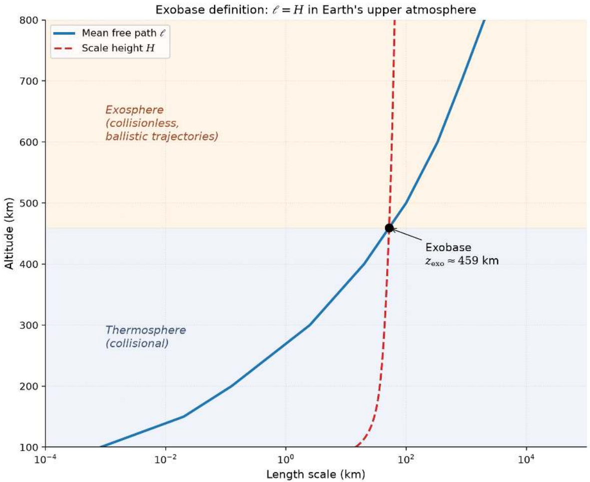
Jeans parameter for Earth and Mars

Because $\lambda_J \propto m$, light hydrogen escapes steadily while heavy molecules remain gravitationally bound.

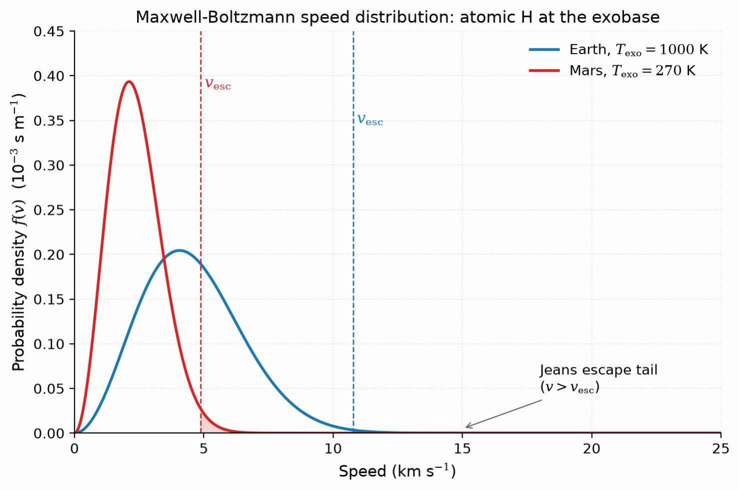
Exobase temperatures: 1000 K (Earth), 270 K (Mars).

Species	m **(u)**	λ_J **(Earth)**	λ_J **(Mars)**
H	1	7.0	5.3
H_2	2	14	11
He	4	28	21
N_2	28	200	150
CO_2	44	310	230

- $\lambda_J \gg 1$ for N_2 and CO_2: thermal escape is completely negligible
- Moderate λ_J allows atomic H to escape steadily from both planets



The exobase: altitude where the mean free path ℓ equals the scale height H, splitting collisional fluid below from ballistic flight above. Course figure.



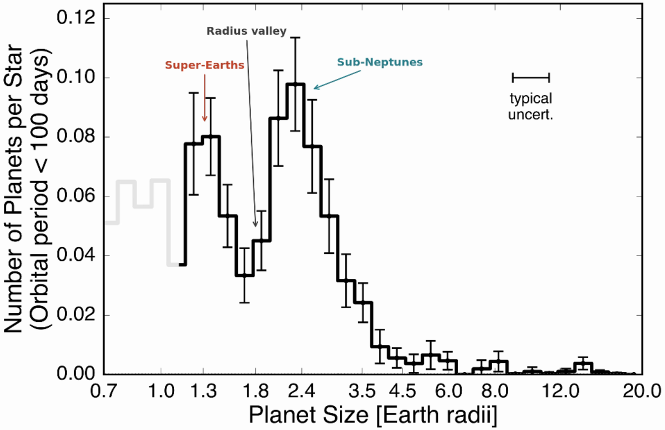
Maxwell-Boltzmann speed distributions for atomic H at Earth and Mars exobases: the escaping fraction sits in the high-velocity tail exceeding v_{esc}, scaling as $e^{-\lambda_J}$. Course figure.

Hydrodynamic escape

Intense stellar EUV heating drives bulk outflow as an **atmospheric wind**:

- Peak importance in first few hundred Myr when stellar EUV is 10–100× higher
- Strips $\mathrm{H_2}$-rich primary envelopes from planets up to several $M_{\oplus}$
- Outflowing light hydrogen can drag along heavier volatiles (He, C, N, O)
- Primary mechanism sculpting the exoplanet radius valley at ~1.5–2 $R_{\oplus}$

Source: Hunten et al. (1987); Owen (2019)



The observed radius valley in the California-Kepler Survey: a bimodal distribution with a deficit near 1.8 $R_\oplus$ separates stripped rocky cores from envelope-retaining sub-Neptunes. After Fulton et al. (2017).

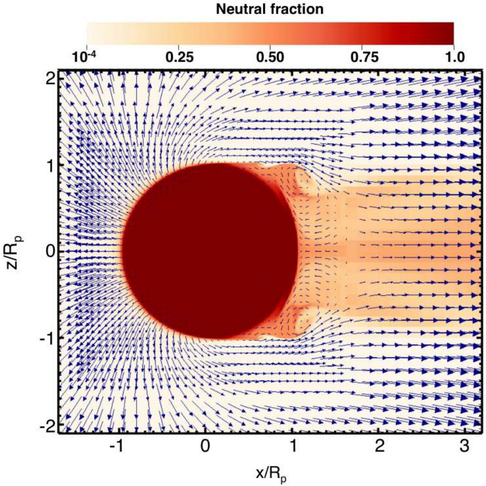
Energy-limited escape rate

Stellar EUV energy absorbed in the upper atmosphere drives hydrodynamic mass loss in the energy-limited regime.

$$\dot{M}_{\mathrm{EL}} = \eta \frac{\pi F_{\mathrm{EUV}} R_p^3}{G M_p}$$

- Efficiency $\eta \approx 0.1$–0.2 converts incident EUV flux into expansion work
- Scales as R_p^3: extended, low-density atmospheres lose mass most rapidly
- Scales inversely with M_p: lower planetary gravity enables stronger bulk outflow

Source: Watson et al. (1981); Owen (2019)



Hydrodynamic outflow from a hot Jupiter driven by stellar EUV heating: heated gas accelerates outward into a night-side wake with Kelvin-Helmholtz shear instabilities. Reproduced from Owen (2019), Fig. 2; simulation from Tripathi et al. (2015).

Non-thermal escape mechanisms

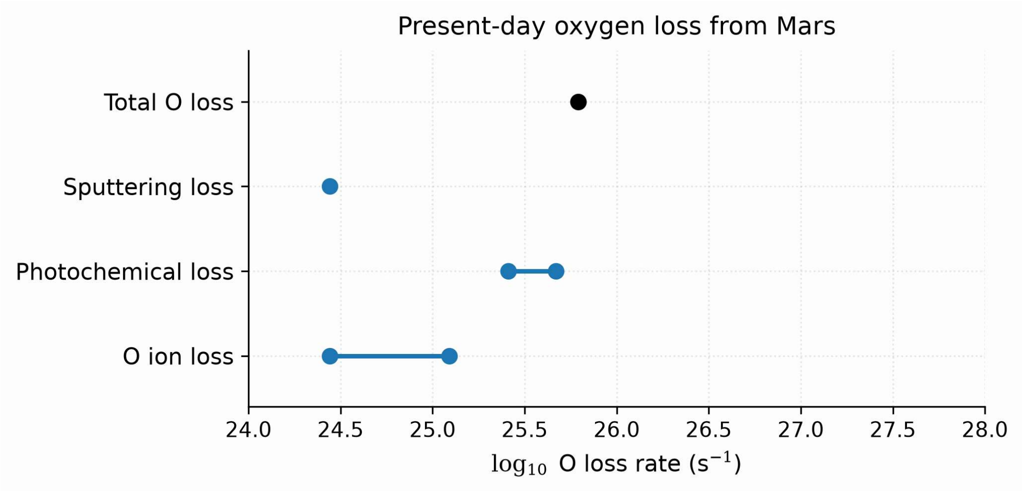
Sputtering: Solar wind ions strike upper atmosphere and eject neutral molecules

Photochemical: Photodissociation yields suprathermal atoms exceeding v_{esc}

Ion pickup: Photoions are swept away by interplanetary magnetic fields

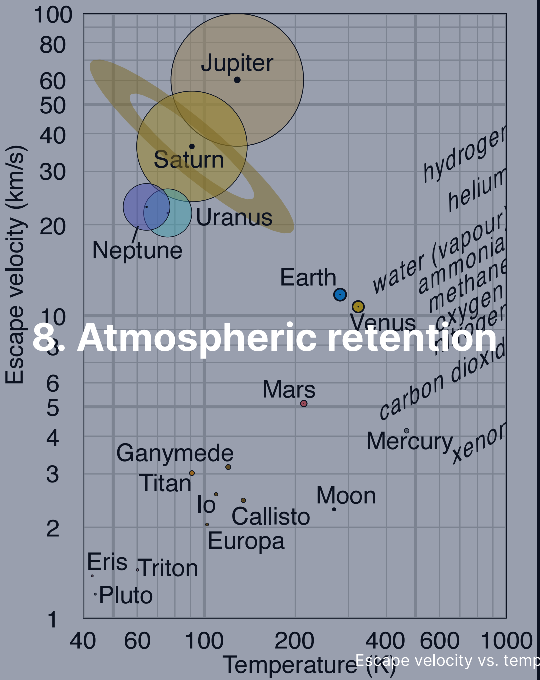
Impact erosion: Giant impacts shock and eject atmospheric columns into space

MAVEN at Mars: photochemical loss of oxygen dominates modern escape over sputtering and ion pickup.



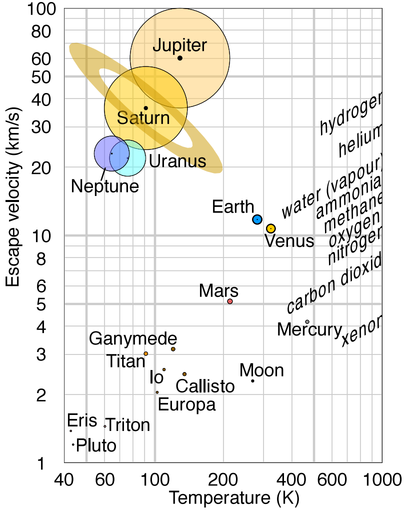
Present-day oxygen loss from Mars by escape channel: photochemical escape dominates modern oxygen loss; summed with hydrogen escape, the total reaches the 2–3 kg s^{-1} MAVEN measures. After Jakosky et al. (2018).

8. Atmospheric retention



Escape velocity vs. temperature. Credit: Wikimedia, CC BY-SA 4.0

Atmospheric retention: v_{esc} vs. T diagram



Credit: Wikimedia, CC BY-SA 4.0

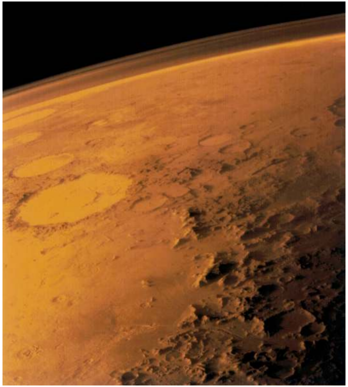
Rule of thumb for long-term retention:

$$v_{\mathrm{esc}} \gtrsim 6\, v_{\mathrm{th}}$$

- Gas giants: retain all species
- Earth/Venus: retain heavy species, lose H
- Mars: marginal; non-thermal losses dominate
- Titan: cold → retains N_2 despite low v_{esc}
- Moon/Mercury: retain no significant atmosphere

Mars: a lost atmosphere

Loss of Mars' magnetic dynamo allowed solar wind stripping to erode an early thick atmosphere down to 6 mbar.

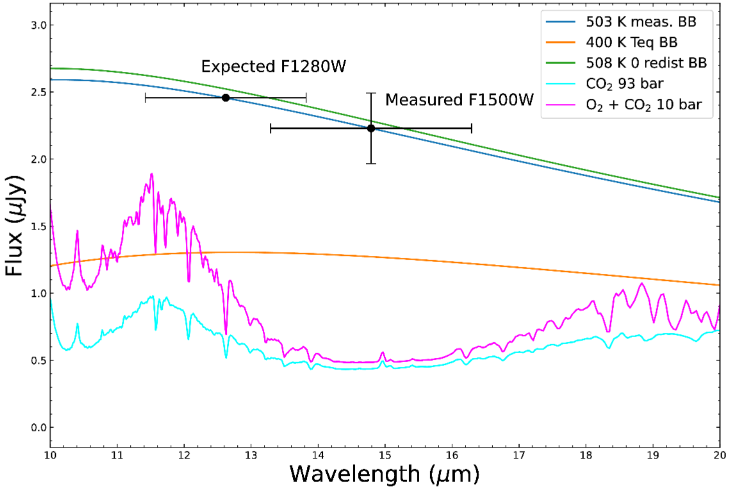


Credit: NASA/JPL (Viking 1, 1976)

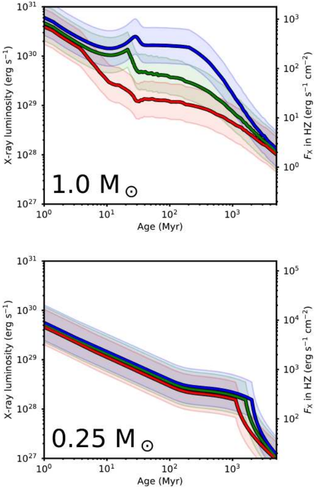
- Today 6 mbar: no stable liquid water, little greenhouse warming
- Valley networks and lakes record a dense early atmosphere
- The dynamo died near ~4.1 Ga and left the atmosphere to the solar wind

9. Recent advances & outlook

Earth’s atmosphere. Credit: NASA/ISS



JWST/MIRI 15 μm thermal emission matches a 503 K bare rock, ruling out a thick atmosphere on TRAPPIST-1 b. Greene et al. (2023).



Stellar X-ray luminosity evolution for Sun-like stars and M dwarfs: lower-mass stars remain in the saturated high-activity regime for over 1 Gyr, driving prolonged atmospheric stripping on close-in rocky exoplanets. Reproduced from Johnstone et al. (2021), Fig. 11.

Summary

1. **Atmosphere types**: primary (captured), secondary (outgassed), tertiary (biogenic)
2. **Hydrostatic balance**: $\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho g$ yields exponential pressure decay $P_0 e^{-z/H}$
3. **Scale height**: $H = k_\mathrm{B} T / (\mu m_u g)$ balances thermal energy against gravity
4. **Vertical structure**: convection sets the lapse rate ($\Gamma_d = g / c_p$ dry, less when condensing) below the 0.1 bar tropopause
5. **Beer-Lambert law**: $I = I_0 e^{-\tau}$; thermal radiation escapes from photosphere at $\tau \approx 1$
6. **Greenhouse warming**: optical depth raises surface temperature T_s above T_eff
7. **Atmospheric escape**: thermal, hydrodynamic, and non-thermal; retention requires $v_\mathrm{esc} \gtrsim 6 v_\mathrm{th}$

Next: Lecture 6

Questions?

Next: Lecture 6, Atmospheres II: clouds, weather & climate (Mara Attia)

Thu 24 Sep, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

`https://ips.formingworlds.space`